Python Visualization Library Documentation

ShadowFox Data Science Internship — Beginner Task

Libraries Covered: Matplotlib & Seaborn

In [1]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd
print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

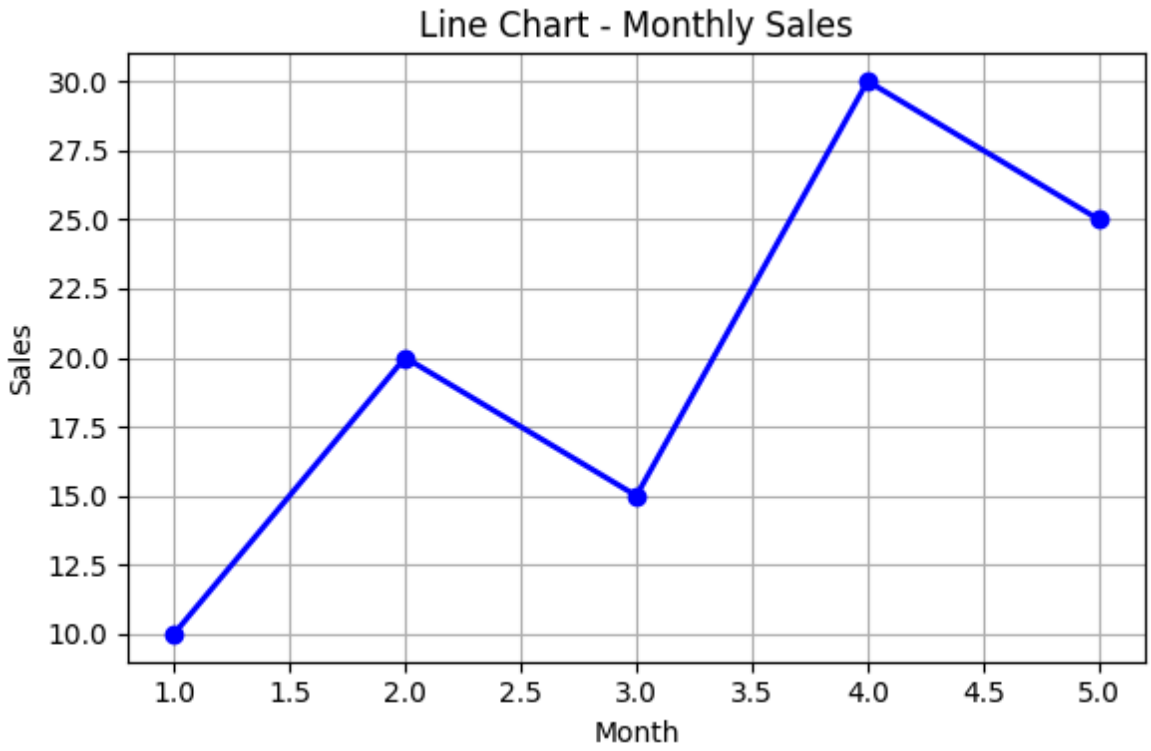
1. MATPLOTLIB

Overview

Matplotlib is a 2D plotting library introduced by John D. Hunter in 2003. Built on NumPy arrays. Foundation for many visualization libraries.

- Use cases: Scientific plots, academic reports, basic data visualization
- Key feature: Full control over every element of a plot

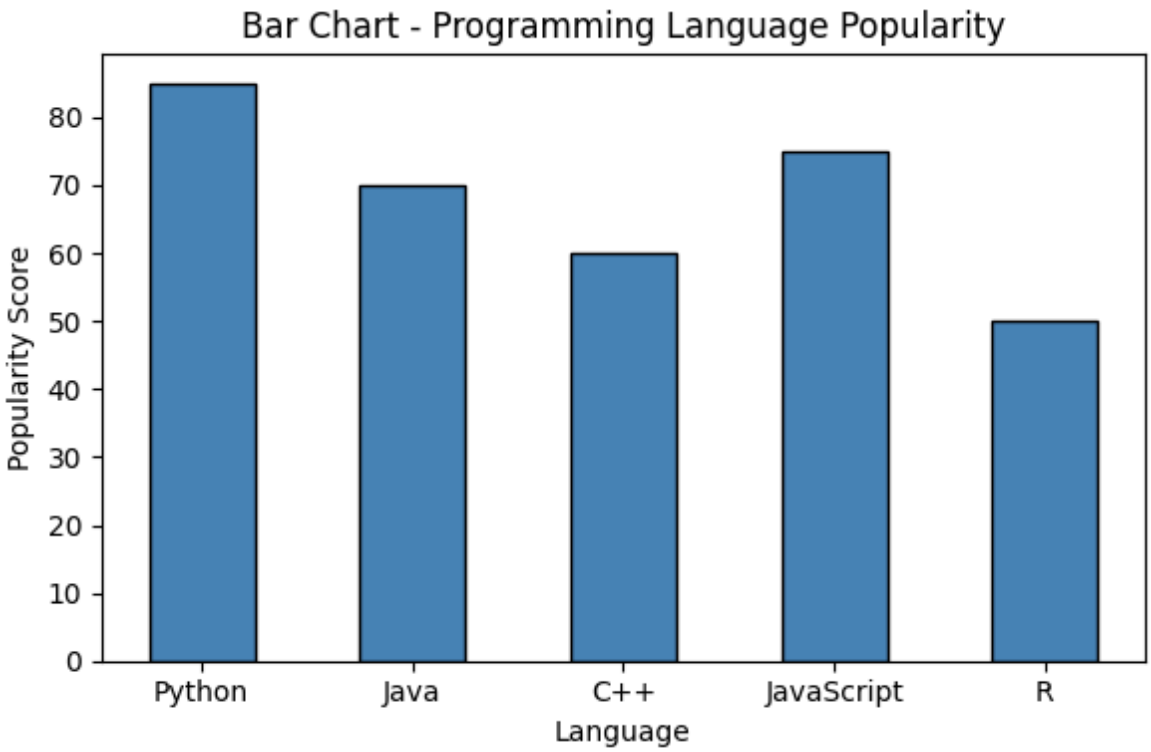
In [2]:
```
x = np.array([1, 2, 3, 4, 5])
y = np.array([10, 20, 15, 30, 25])

plt.figure(figsize=(6, 4))
plt.plot(x, y, color='blue', linewidth=2, marker='o')
plt.title("Line Chart - Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.tight_layout()
plt.show()
print("Use case: Showing trends over time")
```




```
Use case: Showing trends over time
```

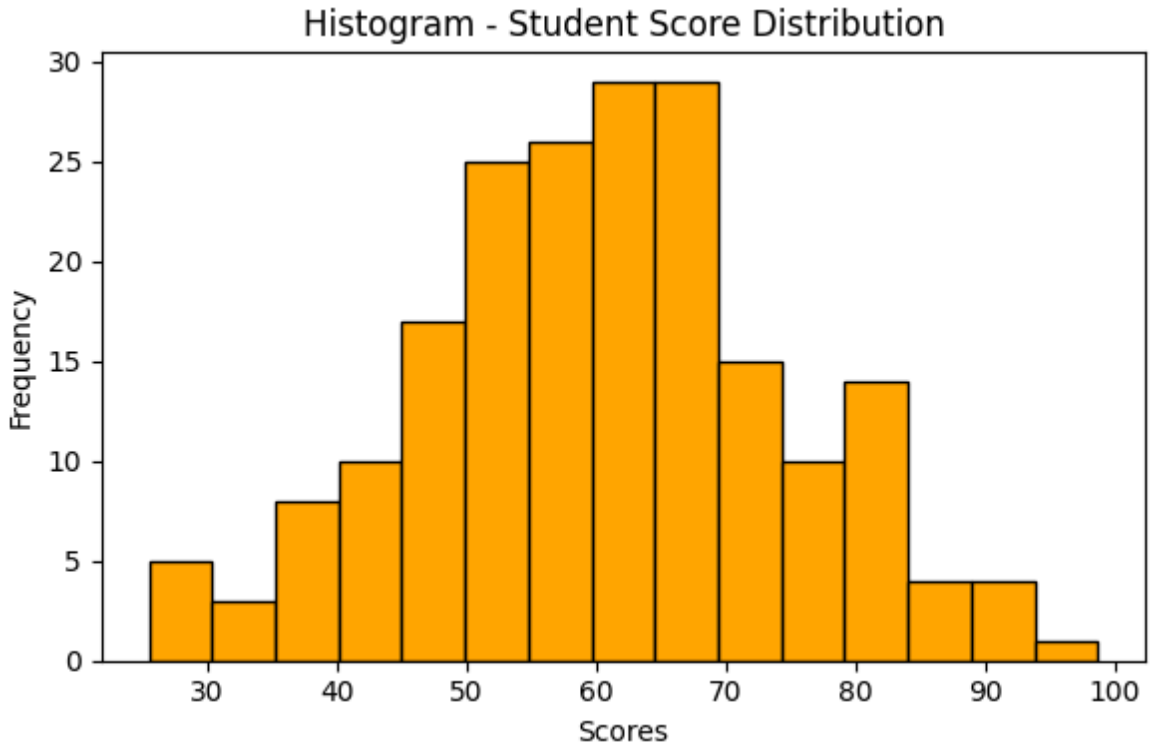
In [3]:

```python
categories = ['Python', 'Java', 'C++', 'JavaScript', 'R']
values = [85, 70, 60, 75, 50]

plt.figure(figsize=(6, 4))
plt.bar(categories, values, color='steelblue', edgecolor='black', width=0.5)
plt.title("Bar Chart - Programming Language Popularity")
plt.xlabel("Language")
plt.ylabel("Popularity Score")
plt.tight_layout()
plt.show()
print("Use case: Comparing categories")
```



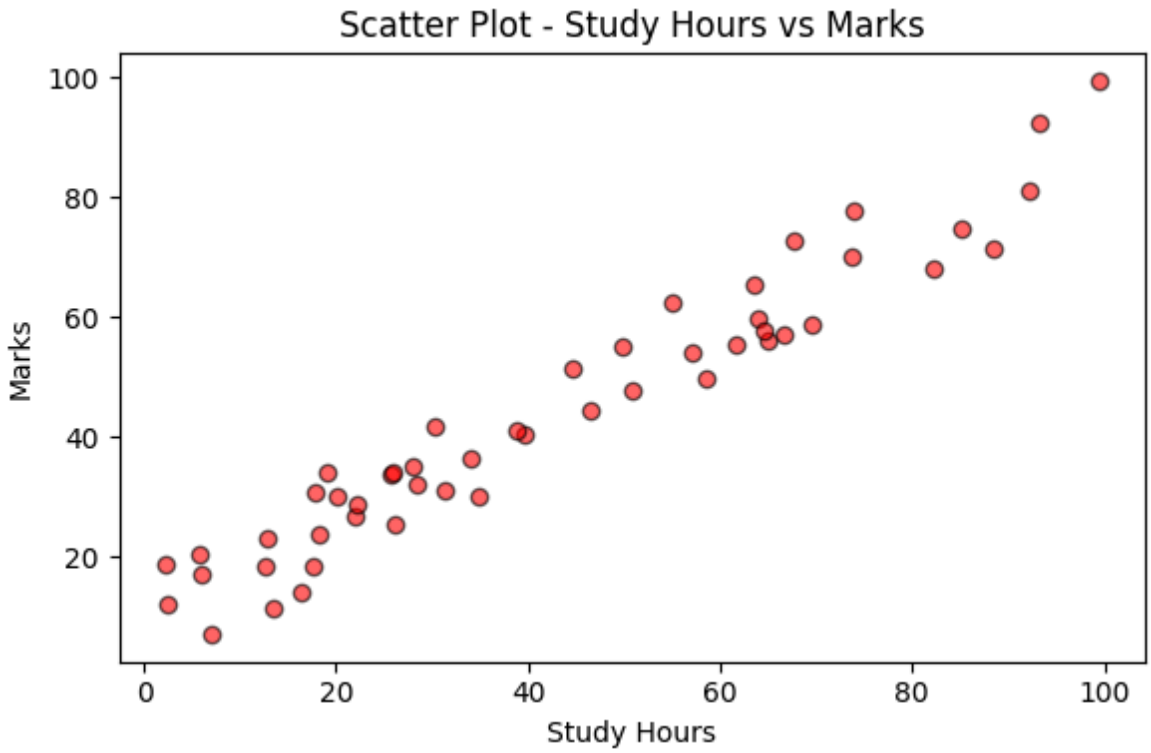
Use case: Comparing categories

In [4]:
```
data = np.random.normal(loc=60, scale=15, size=200)

plt.figure(figsize=(6, 4))
plt.hist(data, bins=15, color='orange', edgecolor='black')
plt.title("Histogram - Student Score Distribution")
plt.xlabel("Scores")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
print("Use case: Frequency distribution of data")
```



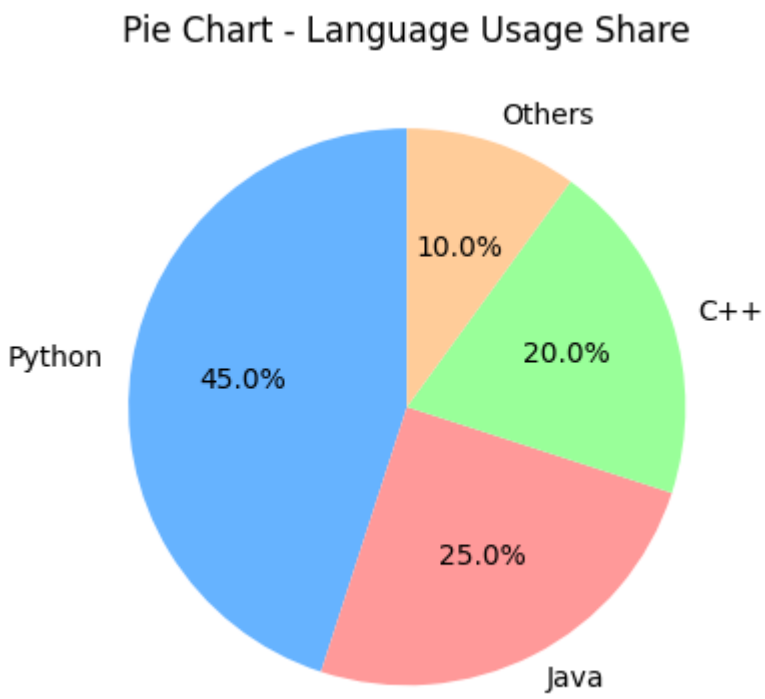
Use case: Frequency distribution of data

In [5]:
```
x = np.random.rand(50) * 100
y = x * 0.8 + np.random.rand(50) * 20

plt.figure(figsize=(6, 4))
plt.scatter(x, y, color='red', alpha=0.6, edgecolors='black')
plt.title("Scatter Plot - Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.tight_layout()
plt.show()
print("Use case: Finding correlation between two variables")
```




```
Use case: Finding correlation between two variables
```

In [6]:
```python
labels = ['Python', 'Java', 'C++', 'Others']
sizes = [45, 25, 20, 10]
colors = ['#66b3ff', '#ff9999', '#99ff99', '#ffcc99']

plt.figure(figsize=(6, 4))
plt.pie(sizes, labels=labels, colors=colors, autopct='%1.1f%%', startangle=90)
plt.title("Pie Chart - Language Usage Share")
plt.tight_layout()
plt.show()
print("Use case: Showing percentage/proportion of categories")
```




```
Use case: Showing percentage/proportion of categories
```

2. SEABORN

Overview

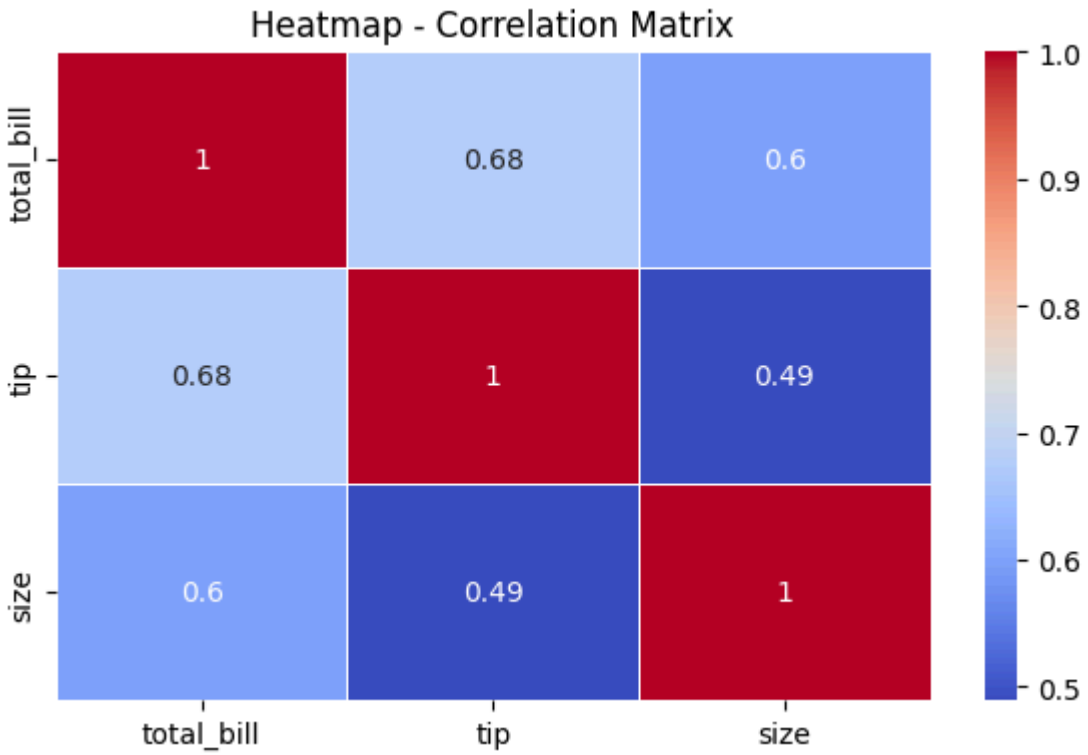
Seaborn is a high-level statistical visualization library built on Matplotlib. Created by Michael Waskom. Produces beautiful plots with minimal code.

- Use cases: Statistical analysis, EDA, correlation analysis
- Key feature: Works directly with Pandas DataFrames

In [7]:
```
tips = sns.load_dataset("tips")
iris = sns.load_dataset("iris")
print("Datasets loaded!")
print(tips.head())
```

```
Datasets loaded!
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
```

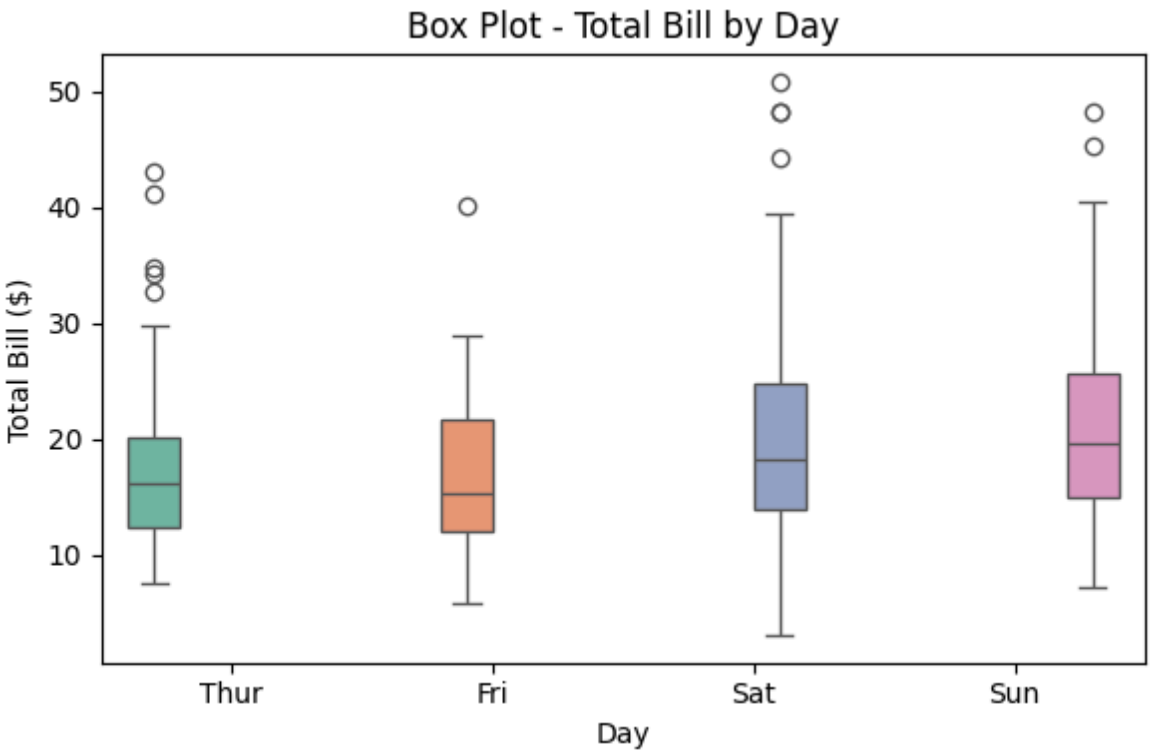
In [8]:
```
plt.figure(figsize=(6, 4))
corr = tips.select_dtypes(include='number').corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title("Heatmap - Correlation Matrix")
plt.tight_layout()
plt.show()
print("Use case: Showing correlation between numerical variables")
```




```
Use case: Showing correlation between numerical variables
```

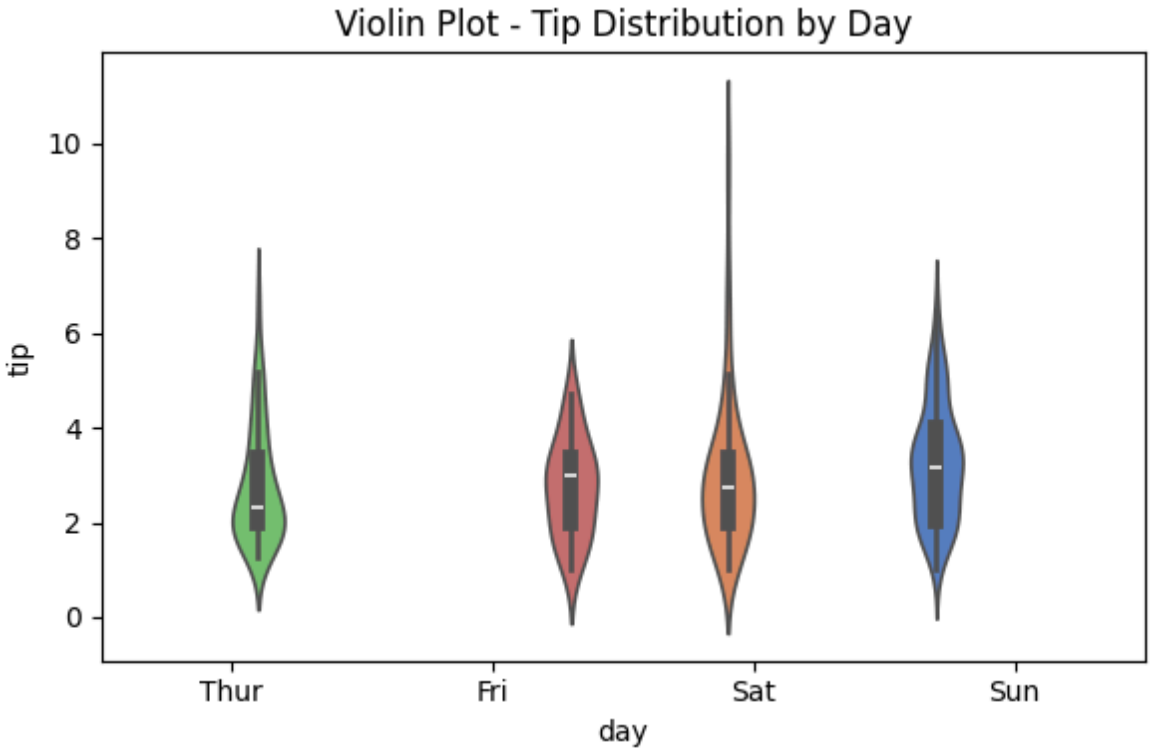
In [9]:

```python
plt.figure(figsize=(6, 4))
sns.boxplot(x='day', y='total_bill', data=tips,
            hue='day', palette='Set2', legend=False)
plt.title("Box Plot - Total Bill by Day")
plt.xlabel("Day")
plt.ylabel("Total Bill ($)")
plt.tight_layout()
plt.show()
print("Use case: Showing data distribution and outliers")
```



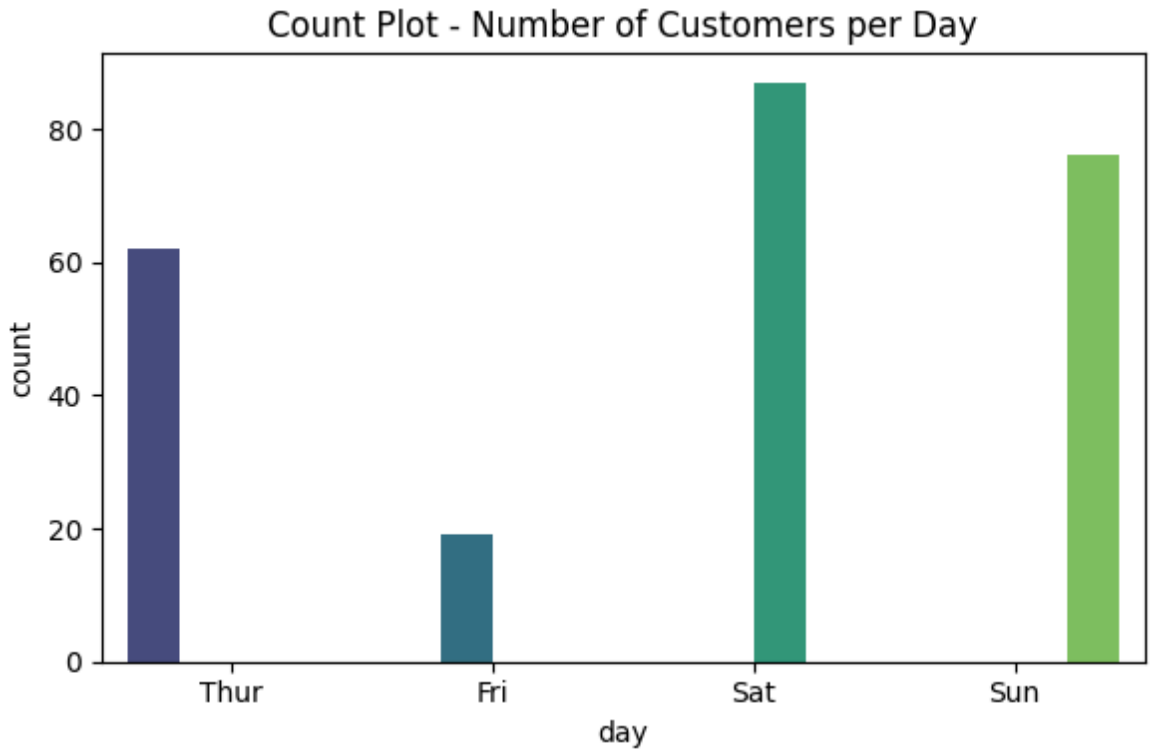

```
Use case: Showing data distribution and outliers
```

In [10]:
```
plt.figure(figsize=(6, 4))
sns.violinplot(x='day', y='tip', data=tips,
               hue='day', palette='muted', legend=False)
plt.title("Violin Plot - Tip Distribution by Day")
plt.tight_layout()
plt.show()
print("Use case: Full distribution shape of data")
```



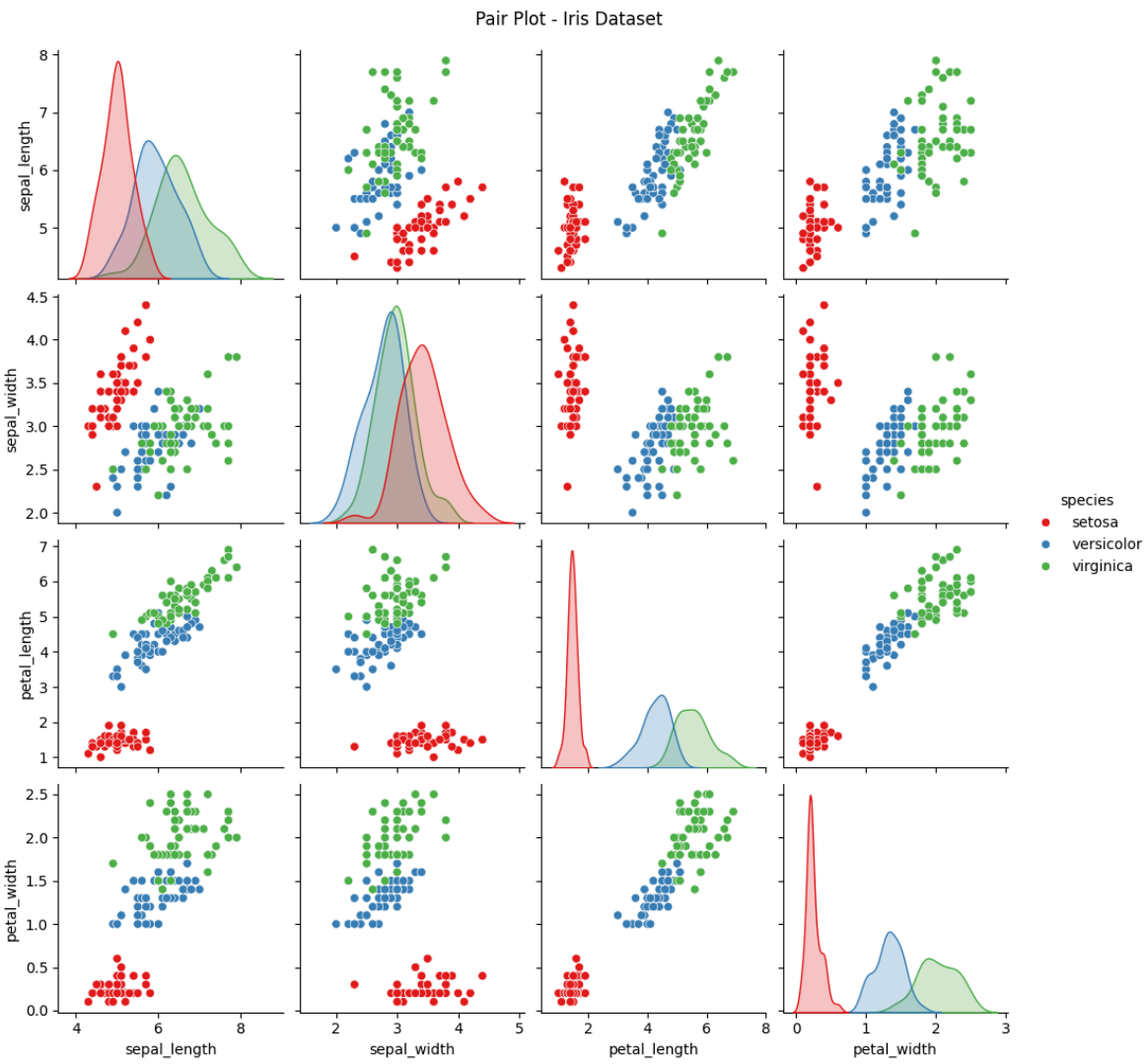

```
Use case: Full distribution shape of data
```

In [11]:
```
plt.figure(figsize=(6, 4))
sns.countplot(x='day', data=tips, hue='day', palette='viridis', legend=False)
plt.title("Count Plot - Number of Customers per Day")
plt.tight_layout()
plt.show()
print("Use case: Count frequency of categorical values")
```




```
Use case: Count frequency of categorical values
```

In [12]:
```
sns.pairplot(iris, hue='species', palette='Set1')
plt.suptitle("Pair Plot - Iris Dataset", y=1.02)
plt.show()
print("Use case: Relationships between all numerical columns")
```




```
Use case: Relationships between all numerical columns
```

3. COMPARISON: Matplotlib vs Seaborn

Feature	Matplotlib	Seaborn
Ease of Use	Medium	Easy
Customization	Very High	Medium
Interactivity	Static	Static
Statistical Plots	Manual	Built-in
Best For	Full control plots	Quick statistical EDA

Conclusion

- Use Matplotlib when you need full control
- Use Seaborn for fast statistical visualization
- Both work best when used together